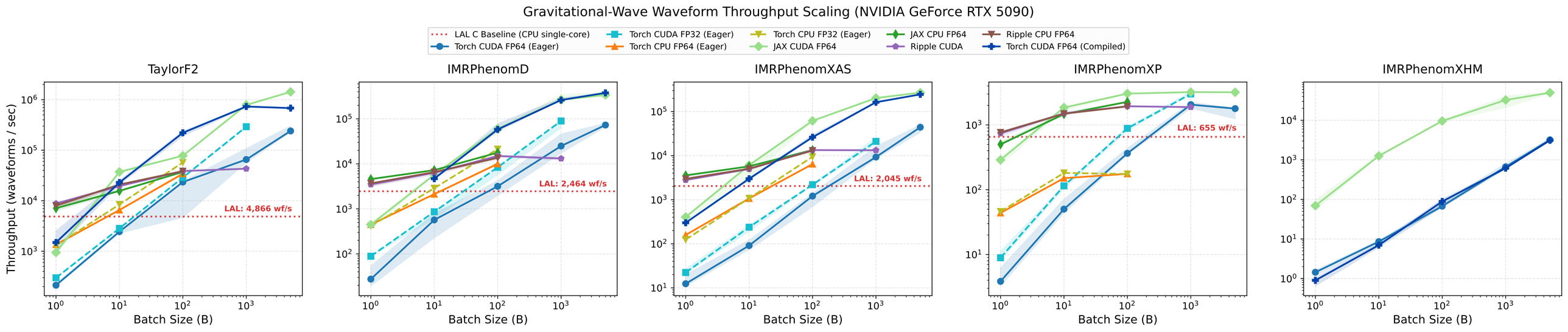
Gravitational-Wave Waveform Throughput Scaling (NVIDIA GeForce RTX 5090)
LAL C Baseline (CPU single-core)
Torch CUDA FP64 (Eager)
Torch CUDA FP32 (Eager)
Torch CPU FP64 (Eager)
Torch CPU FP32 (Eager)
JAX CUDA FP64
JAX CPU FP64
Ripple CUDA
Ripple CPU FP64
Torch CUDA FP64 (Compiled)
TaylorF2
IMRPhenomD
IMRPhenomXAS
IMRPhenomXP
IMRPhenomXHM
Throughput (waveforms / sec)
Batch Size (B)
LAL: 4,866 wf/s
LAL: 2,464 wf/s
LAL: 2,045 wf/s
LAL: 655 wf/s